



Information Technology
PROJECT MANAGEMENT | 7e

Kathy Schwalbe

Chapter 11: Project Risk Management

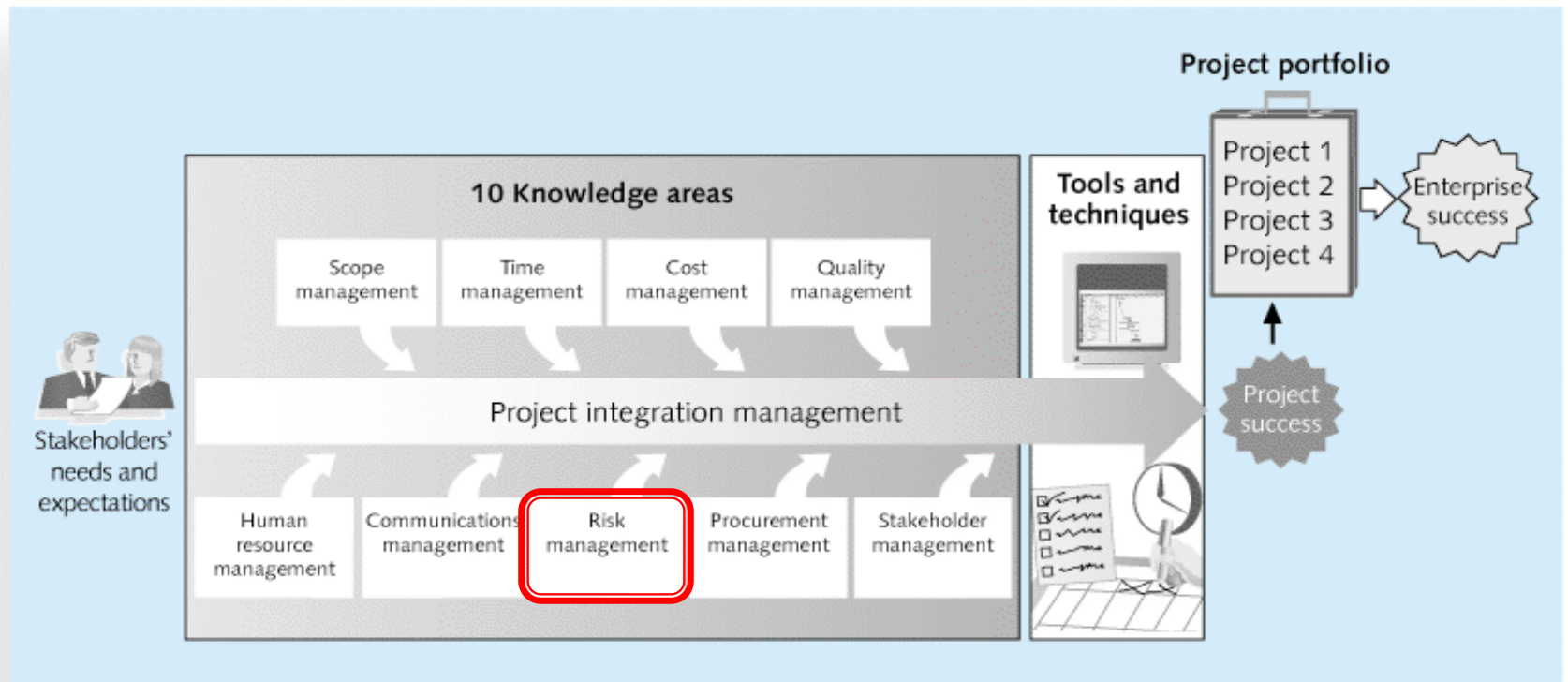
**Information Technology Project
Management, Seventh Edition**

ĐẶNG THỊ THU HÀ – SE
dtthuha79@gmail.com
<https://lms.iuh.edu.vn>



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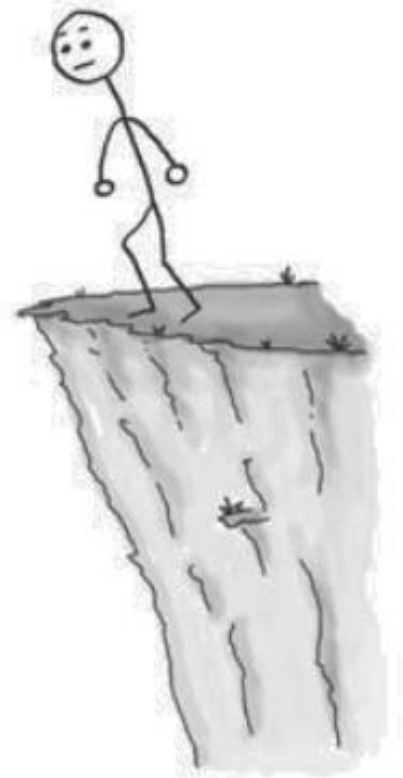


Learning Objectives

- ▶ Explain the concept of risk as it relates to project management, and list the advantages of managing project risks according to best practices
- ▶ Describe the process of identifying risks and create a risk register and risk report
- ▶ Explain quantitative risk analysis and how to apply decision trees, simulation, and sensitivity analysis to quantify risks

The Importance of Project Risk Management

- ▶ Project risk management is the art and science of identifying, analyzing, and responding to risk throughout the life of a project and in the best interests of meeting project objectives
 - Risk management is often overlooked in projects, but it can help improve project success by helping select good projects, determining project scope, and developing realistic estimates



The Process of Project Risk Management

11.1.Planning risk management: deciding how to approach and plan the risk management activities for the project

11.2.Identifying risks: determining which risks are likely to affect a project and documenting the characteristics of each

11.3.Performing qualitative risk analysis: prioritizing risks based on their probability and impact of occurrence

The Process of Project Risk Management

11.4.Performing quantitative risk analysis:

numerically estimating the effects of risks on project objectives

11.5.Planning risk responses: taking steps to enhance opportunities and reduce threats to meeting project objectives

11.6.Monitoring risk: monitoring identified and residual risks, identifying new risks, carrying out risk response plans, and evaluating the effectiveness of risk strategies throughout the life of the project

Planning

Process: **Plan risk management**

Outputs: Risk management plan

Process: **Identify risks**

Outputs: Risk register

Process: **Perform qualitative risk analysis**

Outputs: Project documents updates

Process: **Perform quantitative risk analysis**

Outputs: Project documents updates

Process: **Plan risk responses**

Outputs: Project management plan updates, project documents updates

Monitoring and Controlling

Process: **Control risks**

Outputs: Work performance information, change requests, project management plan updates, project documents updates, organizational process assets updates

Project Start

Project Finish

11.1.Planning Risk Management

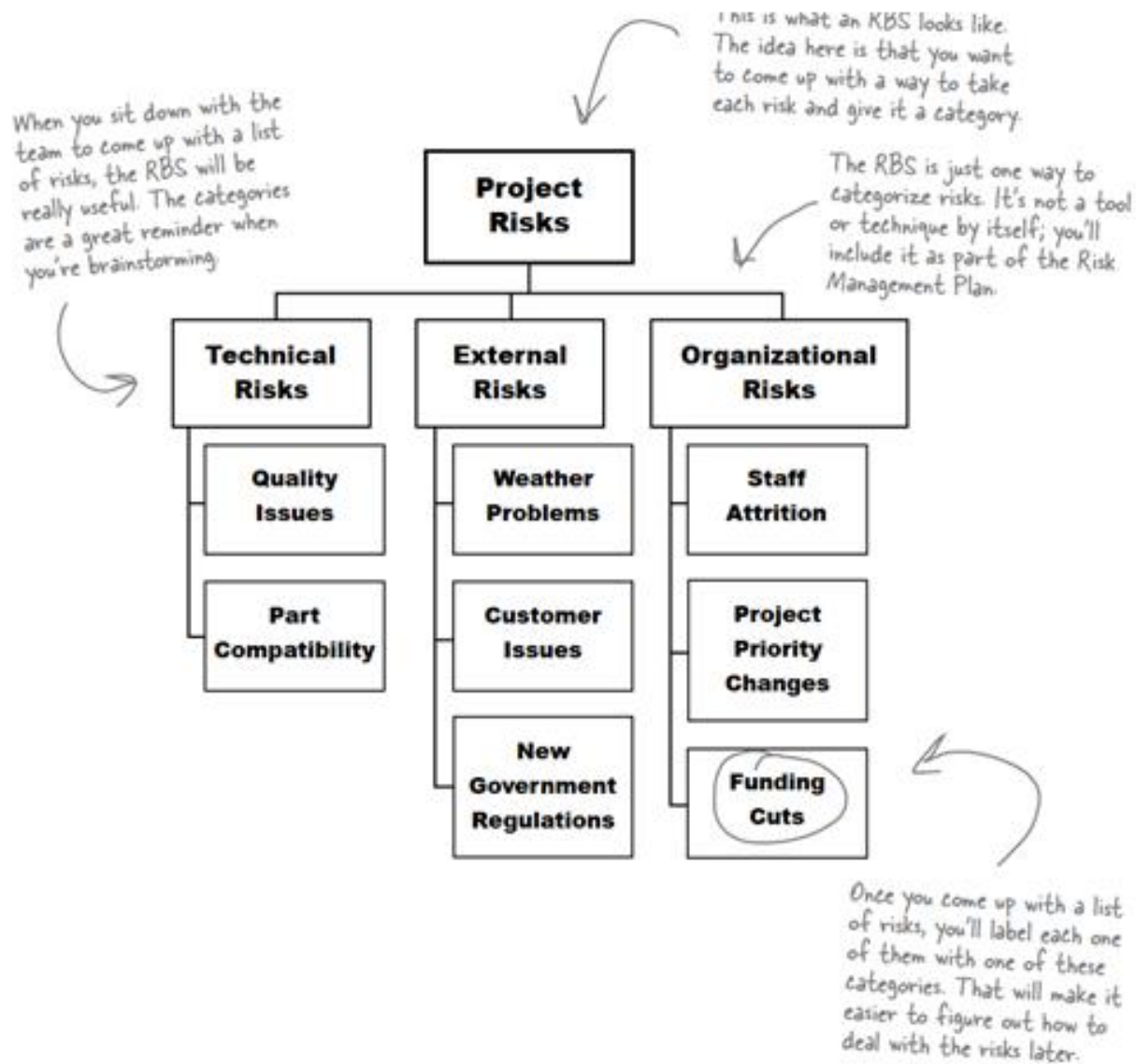
- ▶ Main output of this process is a risk management plan
 - Documents the procedures for managing risk throughout a project
- ▶ The project team should review project documents as well as corporate risk management policies, risk categories, lessons-learned reports from past projects, and templates for creating a risk management plan
 - It is also important to review the risk tolerances of various stakeholders

Topic	Questions to Answer
Methodology	How will risk management be performed on this project? What tools and data sources are available and applicable?
Roles and responsibilities	Which people are responsible for implementing specific tasks and providing deliverables related to risk management?
Budget and schedule	What are the estimated costs and schedules for performing risk-related activities?
Risk categories	What are the main categories of risks that should be addressed on this project? Is there a risk breakdown structure for the project? (See the information on risk breakdown structures later in this chapter.)

Risk probability and impact	How will the probabilities and impacts of risk items be assessed? What scoring and interpretation methods will be used for the qualitative and quantitative analysis of risks? How will the probability and impact matrix be developed?
Revised stakeholders' tolerances	Have stakeholders' tolerances for risk changed? How will those changes affect the project?
Tracking	How will the team track risk management activities? How will lessons learned be documented and shared? How will risk management processes be audited?
Risk documentation	What reporting formats and processes will be used for risk management activities?

Common Sources of Risk on IT Projects

- ▶ Other broad categories of risk help identify potential risks
 - Market risk
 - Financial risk
 - Technology risk
 - People risk
 - Structure/process risk
- ▶ A risk breakdown structure is a hierarchy of potential risk categories for a project



Common Sources of Risk on IT Projects

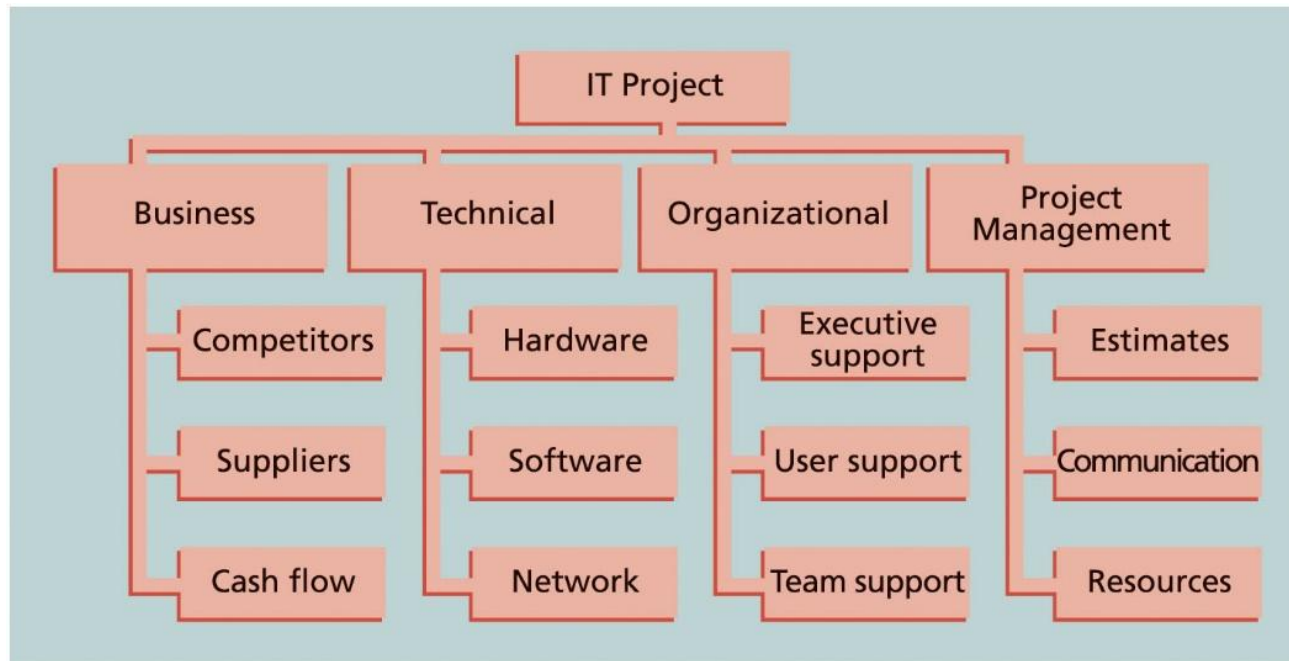


FIGURE 11-4 Sample risk breakdown structure

11.2. Identifying Risks

- ▶ Understanding what potential events might hurt or enhance a particular project
- ▶ Another consideration is the likelihood of advanced discovery
- ▶ Suggestions for identifying risks: tools and techniques
 - Brainstorming
 - The Delphi Technique
 - Interviewing
 - SWOT analysis

▶ Brainstorming

- Group attempts to generate ideas or find a solution for a specific problem by amassing ideas spontaneously and without judgment
- An experienced facilitator should run the brainstorming session
- Be careful not to overuse or misuse brainstorming
 - Psychology literature shows that individuals produce a greater number of ideas working alone than they do through brainstorming in small, face-to-face groups
 - Group effects often inhibit idea generation

▶ Delphi Technique

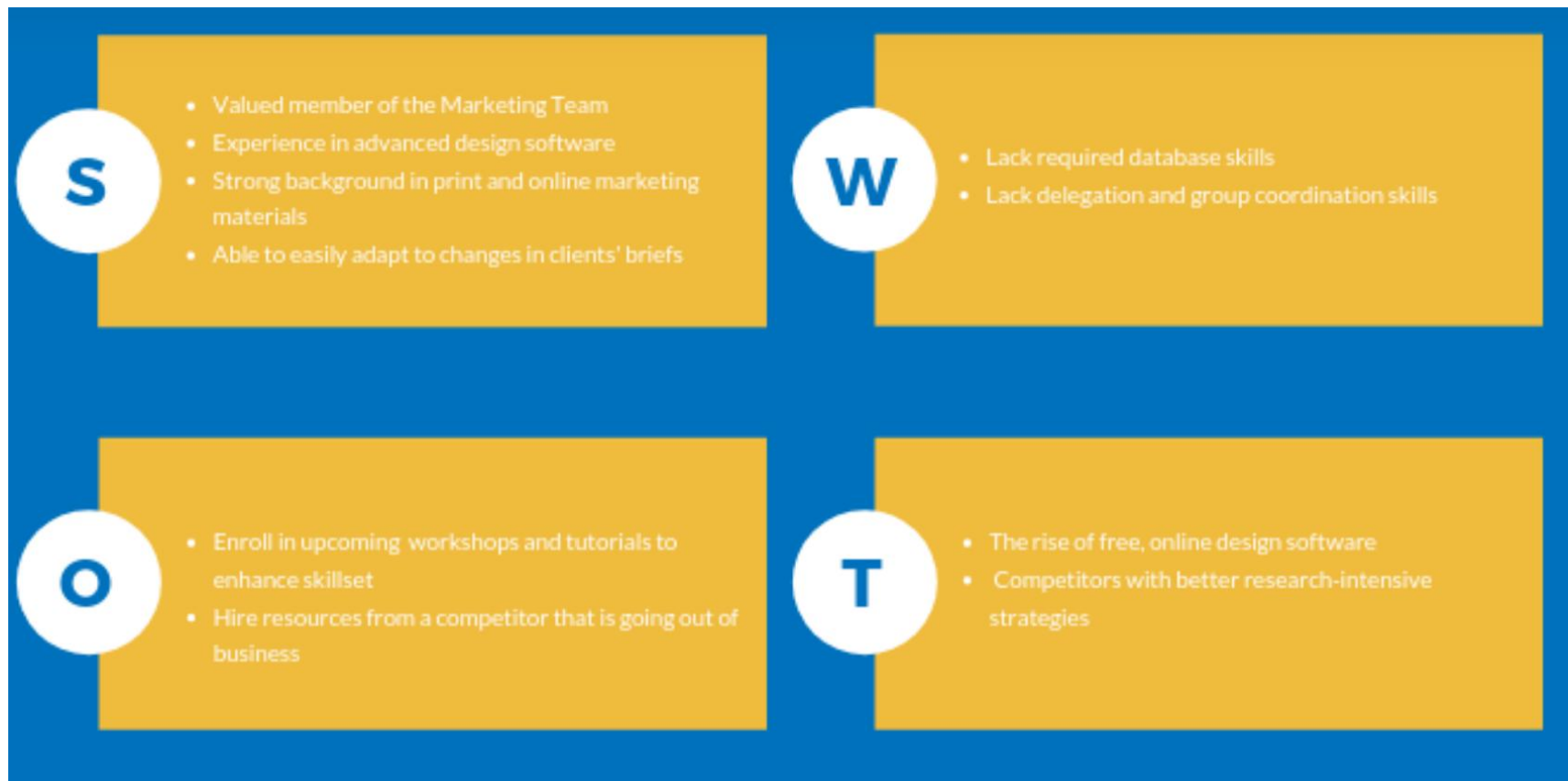
- Used to derive a consensus among a panel of experts who make predictions about future developments
- Provides independent and anonymous input regarding future events
- Uses repeated rounds of questioning and written responses and avoids the biasing effects possible in oral methods

▶ Interviewing

- Fact-finding technique for collecting information in face-to-face, phone, e-mail, or virtual discussions
- Interviewing people with similar project experience is an important tool for identifying potential risks

▶ SWOT analysis

- Strengths, weaknesses, opportunities, and threats
- Helps identify the broad negative and positive risks that apply to a project



Triển khai:

S: Kinh nghiệm...., đi công tác xa,

W: Ko hiểu rõ sản phẩm (function, framework,...),

T: có user không hiểu, gặp trở ngại trên đường di chuyển, ...

The Risk Register

Index	WBS Element	Category	Description	Effects	Expected Effect Date	Probability	Impact	Risk Rank	Owner	Response Plan
0001	1.1.3	Vendors	Web component of the application depends on the design created by the external design team.	Delays in the design delivery will directly impact the project schedule	02.01.2021	6	10	60	D. Nizhebetiski	"Procurement manager should include the following terms in the contract: - Weekly Reports. - Weekly on-site meetings. - Delivery via collaboration system (e.g. Invision)"
0002	1.6.5	HR	Resources for mobile development are limited and on high demand.	- Unavailability of developers may cause delays. - Quality may suffer due to multitasking.	10.02.2021	8	8	64	D. Nizhebetiski	- Develop cross-project HR plan together with Ann Smith and Ron Nagle - Secure required resources.
0003	1.2.3.6	Quality	New functionality may require changes to the infrastructure.	- Poor performance. - Delivery will not be accepted.	01.03.2021	3	10	30	R. Anderson	- Reserve 2 days of work in the beginning of January for Infrastructure Team. - Perform regular performance testing. - Reserve time for performance optimisation (about 2 weeks).
0004	1,3	Scope	Implementation depends on Deliverable 1.1.1.3.	- More efforts will be required to implement the UI.	12.03.2021	3	7	21	J. Robertson	- Task PT-332 was create to incorporate Risk Reserves - BSA should control the vendor on regular basis - Final designs should not require any unsupported technologies.
0005	1.3.2.1	HR	Only one engineer in the company has required skills to deliver the work package. He is currently in conflict with the functional manager. There is a good chance that he will leave the company.	- Delivery will be fully blocked.	01.02.2021	6	10	60	A. Lee E. Remington	- Request HR department to open new vacancy. - Request HR to find an outsourcing/freelancing options. - Start knowledge transfer asap.

Bài Tập

- ▶ Lập bảng danh sách các rủi ro cho bài tập Nhóm (ít nhất 5 rủi ro)
- ▶ Lập bảng phân tích định tính cho danh sách các rủi ro trên

11.3.Performing Qualitative Risk Analysis

- ▶ Assess the likelihood and impact of identified risks to determine their magnitude and priority
- ▶ Risk quantification tools and techniques
 - Probability/impact matrixes
 - Expert judgment

Using Probability/Impact Matrixes to Calculate Risk Factors

- ▶ Lists relative probability of a risk occurring on one side of a matrix or axis on a chart and the relative impact of the risk occurring
 - List the risks and then label each one as high, medium, or low in terms of its probability of occurrence and its impact if it did occur
- ▶ Calculates risk factors
 - Numbers that represent the overall risk of specific events based on their probability of occurring and the consequences to the project if they do occur

Using Probability/Impact Matrixes to Calculate Risk Factors

Probability	High	risk 6	risk 9	risk 1 risk 4
	Medium	risk 3 risk 7	risk 2 risk 5 risk 11	
	Low		risk 8 risk 10	risk 12
		Low	Medium	High
		Impact		

FIGURE 11-5 Sample probability/impact matrix

11.4.Performing Quantitative Risk Analysis

- ▶ Often follows qualitative risk analysis, but both can be done together
 - Large, complex projects involving leading edge technologies often require extensive quantitative risk analysis
- ▶ Main techniques
 - Decision tree analysis
 - Simulation
 - Sensitivity analysis

Decision Trees and Expected Monetary Value (EMV)

- ▶ A decision tree is a diagramming analysis technique used to help select the best course of action in situations in which future outcomes are uncertain
 - Estimated monetary value (EMV) is the product of a risk event probability and the risk event's monetary value
 - You can draw a decision tree to help find the EMV

Decision Trees and Expected Monetary Value (EMV)

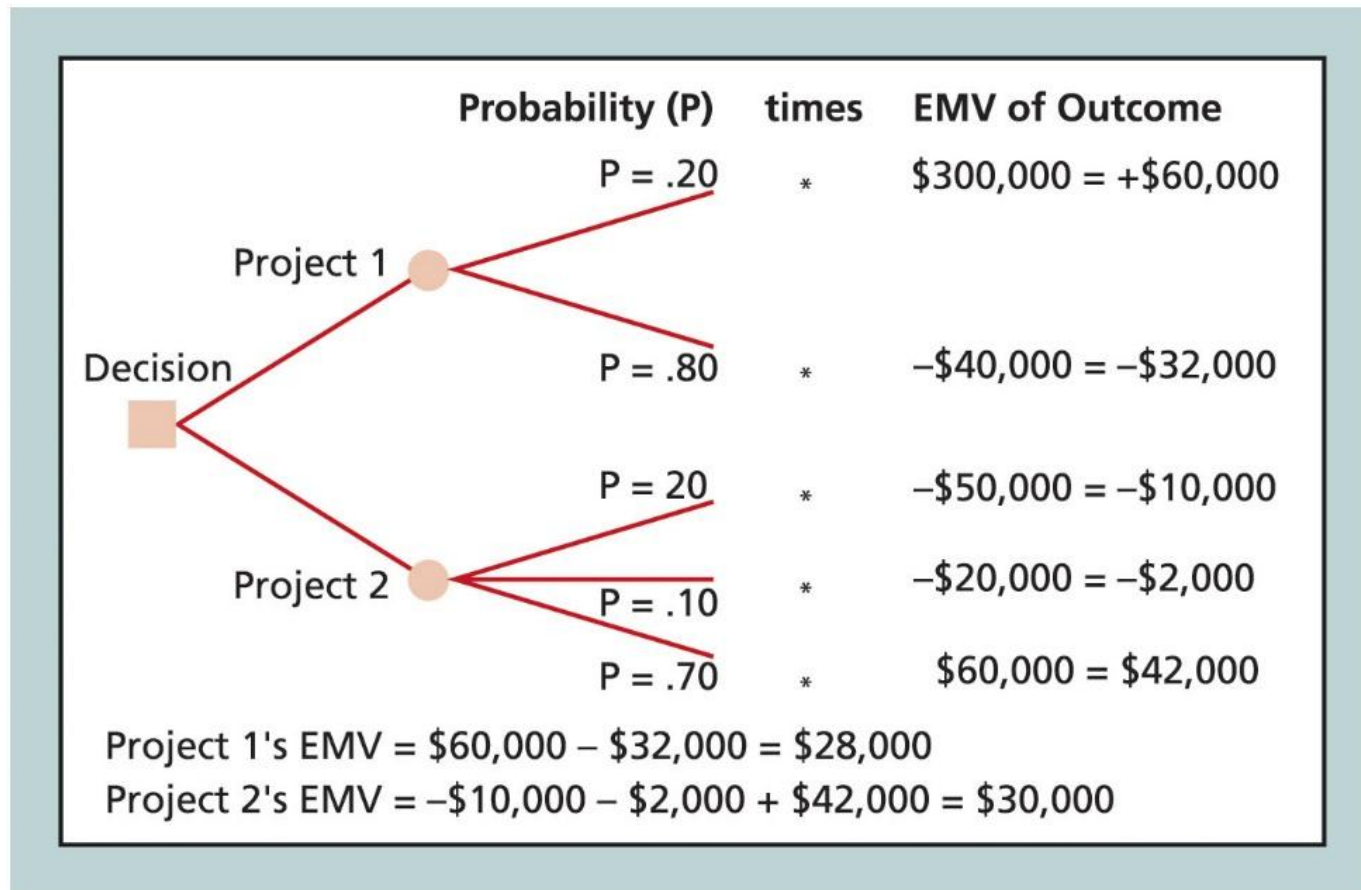


FIGURE 11-7 Expected monetary value (EMV) example

Simulation

- ▶ Uses a representation or model of a system to analyze the expected behavior or performance of the system
 - Monte Carlo analysis simulates a model's outcome many times to provide a statistical distribution of the calculated results
 - Predict the probability of finishing by a certain date or the probability that the cost will be equal to or less than a certain value
 - You can use several different types of distribution functions when performing a Monte Carlo analysis

Simulation

- ▶ Steps of a Monte Carlo analysis
 - Collect the most likely, optimistic, and pessimistic estimates for the variables in the model
 - Determine the probability distribution of each variable
 - Select a random value based on the probability distribution for each variable
 - Run a deterministic analysis or one pass through the model
 - Repeat steps three and four many times to obtain the probability distribution of the model's results

Simulation

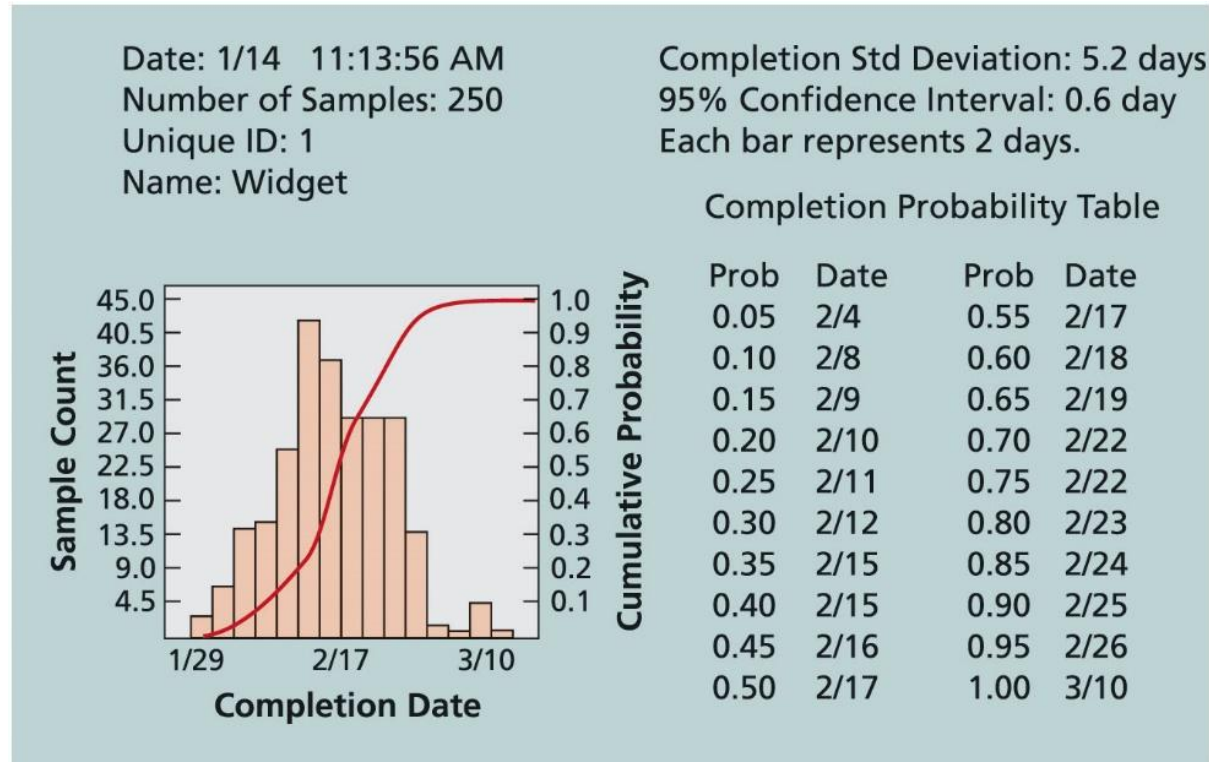


FIGURE 11-8 Sample Monte Carlo-based simulation results for project schedule

Sensitivity Analysis

- ▶ Used to show the effects of changing one or more variables on an outcome
 - For example, many people use it to determine what the monthly payments for a loan will be given different interest rates or periods of the loan
- ▶ Spreadsheet software, such as Microsoft Excel, is a common tool for performing sensitivity analysis

Phân tích rủi ro định tính tập trung vào việc đánh giá từng rủi ro của dự án bằng cách sử dụng xác suất xảy ra và tác động của chúng đến các yếu tố của dự án. kết hợp xác suất và tác động của rủi ro dự án thành điểm rủi ro, sau đó xếp hạng rủi ro để đưa ra mức độ ưu tiên thực hiện hành động.

Dựa vào mức độ rủi ro

Đơn giản thực hiện

Không cần sử dụng phần mềm chuyên dụng

Là đánh giá chủ quan về xác suất và hậu quả

Cần thực hiện đầu tiên

Phân tích ảnh hưởng của rủi ro dự án riêng lẻ

Phù hợp với tất cả dự án với mọi quy mô

Thang điểm và điểm số rủi ro là định tính

Cung cấp thông tin nhanh

Phân tích rủi ro định lượng cung cấp các ước tính số học về mức độ ảnh hưởng của rủi ro đối với các mục tiêu của dự án. Giá trị rủi ro tính toán được so sánh với mức rủi ro chấp nhận được, từ đó sẽ ước tính thời gian và chi phí để hỗ trợ việc ra các quyết định nhằm làm giảm thiểu rủi ro.

Dựa vào mức độ dự án

Tốn nhiều thời gian

Sử dụng phần mềm hỗ trợ trong đánh giá

Là đánh giá khách quan / số học của phân tích xác suất

Cần thực hiện sau khi thực hiện phân tích định tính

Phân tích các tác động kết hợp của rủi ro để xác định rủi ro tổng thể của dự án

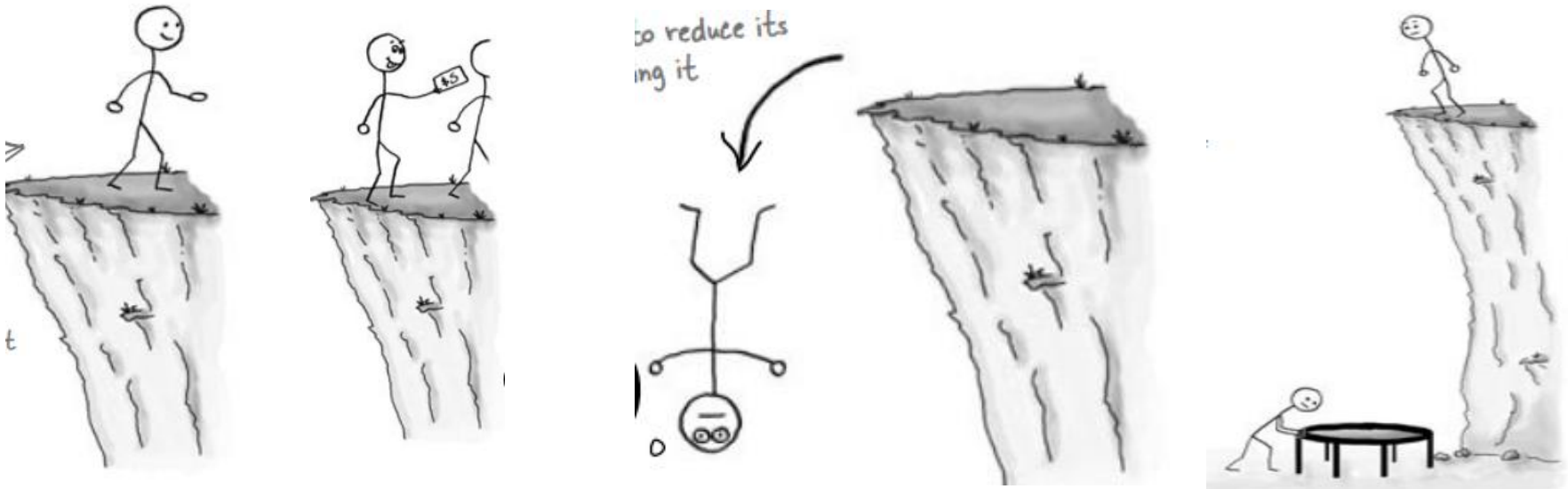
Đặc biệt phù hợp trong các dự án lớn và phức tạp

Thang điểm và điểm số rủi ro là định lượng

Cung cấp thông tin chi tiết

11.5.Planning Risk Responses

- Risk avoidance
- Risk transference
- Risk acceptance
- Risk mitigation



11.6. Monitoring Risks

- ▶ Involves ensuring the appropriate risk responses are performed, tracking identified risks, identifying and analyzing new risk, and evaluating effectiveness of risk management throughout the entire project
 - Project risk management does not stop with the initial risk analysis

Monitoring Risks

- ▶ Carrying out individual risk management plans involves monitoring risks based on defined milestones and making decisions regarding risks and their response strategies
 - Project teams sometimes use workarounds—unplanned responses to risk events—when they do not have contingency plans in place

Chapter Summary

- ▶ Risk is an uncertainty that can have a negative or positive effect on meeting project objectives
- ▶ Risk management is an investment
- ▶ Implementing risk responses involves putting the appropriate risk response plans into action
- ▶ Monitoring risks involves monitoring implementation of risk response plans, tracking identified risks, identifying and analyzing new risks, and evaluating effectiveness of risk management throughout the entire project